

Foundations of Trustworthy AI

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CS7180/7880 · Fall 2026

Class 7: Wrap Calibration → Introduce Robustness

Oct 2, 2026

Roadmap

Omniprediction
Connection to Group Calibration

Entering
Robustness

Overview of Robustness
Examples

Conformal Prediction
Marginal vs. Group Coverage

Recap Definitions

Definition: For distribution D , predictor f , group class G , f is α -group calibrated if

$$\forall g \in G \sum_{p \in \text{range}(f)} \beta(f, g_{f,p})^2 \cdot \mu(g_{f,p}) \leq \alpha^2$$

Definition: For distribution D , actions A , and collection of losses $\mathcal{L}$, and collection of predictors $\mathcal{H} = \{h: X \rightarrow [0,1]\}$, f is an α -omnipredictor if for every $l \in \mathcal{L}$, every $h \in \mathcal{H}$

$$\mathbb{E}_D[l(\underline{\text{BR}}_l(f(x)), y)] \leq \mathbb{E}_D[l(\text{BR}_l(h(x)), y)] + \alpha$$

Best response assuming p is true probability

Tangent #1

$$\begin{array}{ccc} f(x) & \longmapsto & f(x) + \sum_{g \in \mathbb{G}} \beta_g g(x) \\ \in \mathcal{H} & & \in \mathcal{H} \end{array}$$

Calibration $\rightarrow$ Omniprediction

$$x \xrightarrow{h} p \xrightarrow{BR_e} a$$

$$g_{h,\ell,a}(x) := \mathbb{1}\{BR_\ell(h(x)) = a\} \quad h \in \mathcal{H}, \ell \in \mathcal{L}, a \in A$$

$$\mathcal{G}_{\mathcal{H},\mathcal{L}} := \{g_{h,\ell,a} : h \in \mathcal{H}, \ell \in \mathcal{L}, a \in A\}$$

Class of groups we
care about

Theorem: If f is (perfectly) calibrated for $\mathcal{G}_{\mathcal{H},\mathcal{L}}$ then
 f is a (perfect) omnipredictor for $(\mathcal{H}, \mathcal{L})$

Calibration $\rightarrow$ Omniprediction

Theorem: If f is (perfectly) calibrated for $G_{H, \mathcal{L}}$ then f is a (perfect) omnipredictor for $(H, \mathcal{L})$

Proof: Fix $\ell \in \mathcal{L}$, $h \in H$, define $A = \text{BR}_\ell(h(X))$

Calibration $\Rightarrow \mathbb{P}[Y=1 \mid f(X)=p, \text{BR}_\ell(h(X))=a] = p$

$$\begin{aligned} & \mathbb{E}_{(X,Y) \sim D} [\ell(\text{BR}_\ell(f(X)), Y) \mid f(X)=p, \text{BR}_\ell(h(X))=a] \\ &= \mathbb{E}_{Y \sim \text{Ber}(p)} [\ell(\text{BR}_\ell(p), Y)] \leq \mathbb{E}_{Y \sim \text{Ber}(p)} [\ell(a, Y)] \\ &= \mathbb{E}_{(X,Y) \sim D} [\ell(\text{BR}_\ell(h(X)), Y) \mid \text{BR}_\ell(h(X))=a, f(X)=p] \end{aligned}$$

Conformal Prediction

$\mathcal{X}$ is a feature space

D is a distribution on (X, Y)

$\mathcal{Y}$ is a label/outcome space

A prediction set is a function $T: \mathcal{X} \rightarrow 2^{\mathcal{Y}}$

Goal: Design prediction sets with "good coverage"

Pointwise coverage: for every $x \in \mathcal{X}$, $\mathbb{P}_{(X,Y)}[Y \in T(x) | X=x] \geq 1 - \delta$

Average coverage: $\mathbb{P}_{(X,Y) \sim D} [Y \in T(x)] \geq 1 - \delta$

Conformal Prediction

Score: $s: \mathcal{X} \times \mathcal{Y} \rightarrow \mathbb{R}$

Examples:

- Regression: given a predictor $f: \mathcal{X} \rightarrow \mathbb{R}$

$$s(x, y) = (f(x) - y)^2$$

- Multiclass classification: given a predictor $f: \mathcal{X} \rightarrow \Delta(\mathcal{Y})$

$$s(x, y) = 1 - f(x)_y$$

Prediction Set:

$$T_{s, \tau}(x) = \{ y \in \mathcal{Y} : s(x, y) \leq \tau \}$$

Split Conformal Prediction

- s is fixed
- draw a calibration set $(X_1, Y_1), (X_2, Y_2), \dots, (X_n, Y_n) \sim D$
- choose a parameter k
 - test point (X_0, Y_0)

- 1: Let $S_i = s(X_i, Y_i)$ for $i \in [n]$
- 2: Let $\hat{\tau}$ be the k^{th} smallest of $S_1, \dots, S_n$
- 3: $T_{s, \hat{\tau}}(x) = \{y \in Y : s(x, y) \leq \hat{\tau}\}$

Theorem:
$$\mathbb{P} \left[Y_0 \in T_{s, \hat{\tau}_{(k)}}(X_0) \right] = \frac{k}{n+1}$$

$\begin{matrix} (X_1, Y_1) \\ \vdots \\ (X_n, Y_n) \end{matrix} \quad \begin{matrix} (X_0, Y_0) \\ \text{calibration data} \end{matrix}$

$S_0 = s(X_0, Y_0), S_1 = s(X_1, Y_1), \dots, S_n = s(X_n, Y_n)$ ($n+1$ iid scores)

S_0 is k^{th} smallest
or smaller w.p. $k/n+1$

